

Genetic Syndrome Classification from Image Embeddings

1. Introduction

This report details the methodology, results, and insights derived from an embedding classification pipeline designed to identify various genetic syndromes. The primary objective is to classify these syndromes utilizing 320-dimensional image embeddings extracted from a pre-trained model. Furthermore, the project aims to analyze the underlying data distribution and systematically evaluate classification performance by employing the K-Nearest Neighbors (KNN) algorithm across different distance metrics.

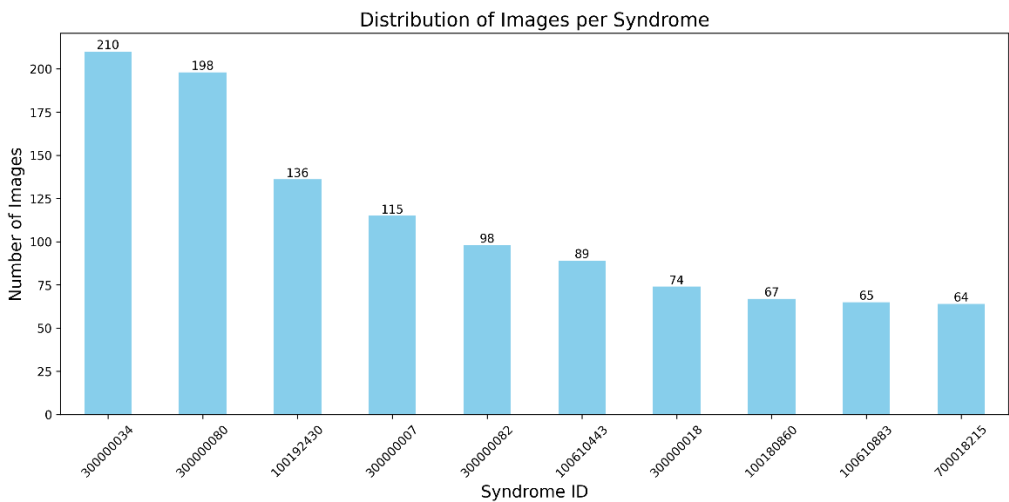
2. Methodology

2.1 Data Processing

Data loading and preprocessing were implemented in `processing.py`. The dataset was retrieved from the provided pickle file, which contained a hierarchical structure organized by `syndrome_id`, `subject_id`, and `image_id`. This structure was flattened into a tabular format suitable for analysis and model training. During this transformation process, strict data integrity checks were enforced; consequently, any missing values, invalid entries, or embeddings lacking the expected 320 dimensions were identified and removed.

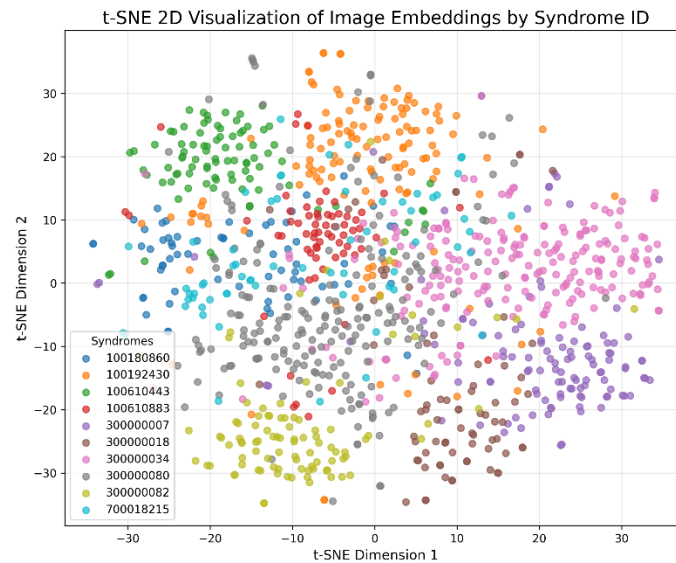
2.2 Exploratory Data Analysis

The following figure presents the number of valid image embeddings for each syndrome IDs in the dataset:



The distribution reveals a significant class imbalance across syndromes. The most represented class (ID: 300000034) contains 210 images, which is more than triple the volume of data available for the least represented class within this top-tier subset (e.g., ID: 700018215 with 64 images). This pronounced skewness indicates that the predictive model will have substantially more exposure to the feature spaces of the top two or three syndromes during the training phase. Consequently, a baseline K-Nearest Neighbors (KNN) classifier might become biased toward predicting the majority classes.

To better understand the structure of the embedding space, t-Distributed Stochastic Neighbor Embedding (t-SNE) was applied to reduce the 320-dimensional vectors into a two-dimensional representation. The figure below presents the 2D projection of the embeddings.



The 2D t-SNE visualization reveals complex relationships between the genetic syndrome embeddings. For instance, syndromes 100610443 (green), 300000082 (yellow), 300000018 (brown), and 300000007 (purple) are clearly separated in specific regions of the 2D space, indicating strong intra-class similarity and good embedding coherence. Conversely, a significant degree of overlap exists in the central region of the plot, where syndromes such as 300000080 (grey), 100610883 (red), and 300000034 (pink) exhibit significant spatial intersection. This suggests that their embeddings share similar feature representations.

In such scenarios, model performance becomes sensitive to the choice of distance metric (Euclidean versus Cosine) and the selected number of neighbors (k). Moreover, the combined effects of class imbalance and cluster overlap make overall accuracy insufficient as a sole evaluation criterion. Macro-averaged F1-score, Averaged Area Under the Curve (AUC), and Top-k accuracy provide a more reliable assessment of performance across all classes. This also reinforces the importance of stratified cross-validation to preserve class proportions and ensure fair evaluation.

2.3. Classification Task

The classification task was addressed using KNN algorithm. KNN is a distance-based method that assigns a class label to a sample based on the majority label among its closest neighbors in the embedding space.

A critical consideration during this phase was the prevention of data leakage. Because the dataset contains multiple images per subject (``subject_id``), utilizing a standard random split could inadvertently place embeddings from the same individual into both the training and testing sets. To resolve this, ``StratifiedGroupKFold`` was utilized for the cross-validation splitting process. This approach ensures two crucial conditions are met simultaneously:

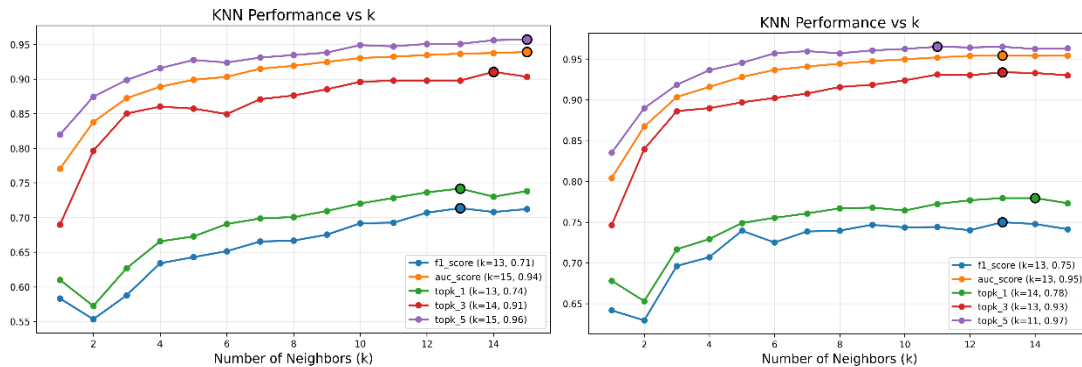
- All images belonging to the same subject are kept within the same fold. This prevents data leakage.
- Each fold preserves the class distribution, mitigating bias caused by class imbalance.

Grouping constraints are particularly important in medical datasets, where multiple images of the same subject may share highly similar embedding representations. Without grouping, model performance could be artificially inflated due to memorization of subject-specific patterns.

3. Results

To ensure robust and unbiased evaluation, a 10-fold StratifiedGroupKFold strategy was adopted. The classification pipeline systematically evaluated the KNN model using neighbor values k ranging from 1 to 15. Furthermore, to determine the most effective method for measuring embedding similarity in a high-dimensional space, the algorithm was tested using two distinct distance metrics: Euclidean and Cosine.

Performance metrics were computed per fold, and results were averaged across all 10 folds. The figures below summarize model performance as a function of k . The left plot corresponds to Euclidean distance, while the right plot presents results using Cosine distance.

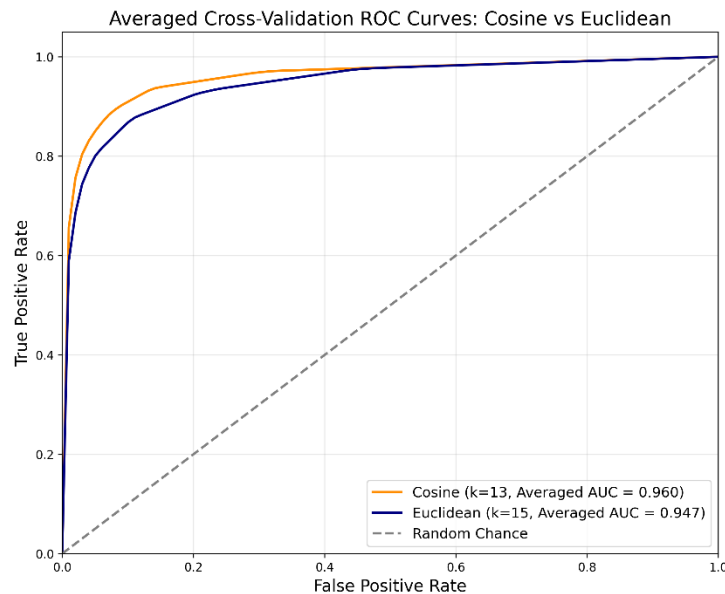


Across both distance metrics, performance improves substantially when increasing k from 1 to approximately 8 neighbors. Small values of k produce higher variance and lower stability, particularly for F1-score and Top 1 accuracy. As k increases, performance stabilizes and gradually reaches a plateau, indicating that incorporating more neighbors reduces sensitivity to local noise.

The figure shows that Cosine distance consistently outperforms Euclidean distance across all evaluated metrics. For instance, the best F1-score with Cosine distance approaches approximately 0.75, while Euclidean remains slightly lower (0.71). Similar improvements are observed for top-k and AUC.

In high-dimensional spaces (such as the 320-dimensional embeddings used here), the Euclidean distance becomes highly sensitive to the magnitude of the vectors, which can be skewed by scaling differences in the deep learning model's output. Cosine distance, however, measures the angle between vectors, focusing purely on the orientation and pattern of the features rather than their absolute magnitude. This makes Cosine similarity significantly more robust and effective for capturing the semantic relationships between high-dimensional image embeddings.

To further evaluate classifier performance beyond threshold-dependent metrics, Receiver Operating Characteristic (ROC) curves were computed using a One-vs-Rest strategy and averaged across the 10 cross-validation folds. The figure below compares the average ROC curves obtained using Cosine and Euclidean distance metrics with their respective optimal values of k .



When comparing the two metrics, the Cosine distance (evaluated at its optimal $k=13$) exhibits superior performance, achieving an AUC of 0.960. In contrast, the Euclidean distance (evaluated at its optimal $k=15$) achieves a slightly lower average AUC of 0.947.

Visually, the orange curve representing the Cosine metric climbs more steeply toward the top-left corner of the plot. This indicates that the Cosine-based classifier can achieve higher sensitivity (identifying true positive cases of a syndrome) with fewer false alarms compared to the Euclidean approach. This is particularly relevant in medical applications, where minimizing false positives while maintaining high sensitivity is often critical. This finding consistently reinforces the earlier conclusion: Cosine similarity is a more robust measure for high-dimensional image embeddings, as it effectively captures the semantic direction of the feature vectors while mitigating the impact of magnitude variance.

The experimental results demonstrate a clear relationship between embedding structure and classification performance. Syndromes ID forming compact and well-separated clusters achieve higher accuracy, while overlapping regions in the embedding space lead to reduced discriminability and lower Top-1 accuracy.

Overall, the findings confirm that cosine-based similarity consistently outperforms Euclidean distance, indicating that directional relationships within the embedding space better capture syndrome-specific characteristics in this dataset.

4. Challenges and Solutions

- **Data Structure Comprehension.** The provided pickle file (`mini\_gm\_public\_v0.1.p`) did not include metadata or detailed documentation regarding its internal schema.

Solution: To overcome this, the loaded file was systematically explored by printing its contents and inspecting the nested data types. This step was essential for designing the flattening algorithm used in the preprocessing stage.

- **F1-Score, ROC/AUC and Top-k metrics:** The correct interpretation of these metrics for a multiclass problem.

Solution: The foundational concepts of the binary confusion matrix—specifically True Positives (TP), False Positives (FP), and False Negatives (FN)—were thoroughly reviewed. These concepts were then extended to the multiclass domain using a One-vs-Rest approach. The precision and recall were calculated for each individual class, and a macro-average was computed to yield the final F1-Score, ensuring that the heavily underrepresented classes were weighted equally against the majority classes.

4. Recommendations

- **Multimodal Feature Integration:** To provide more context to the classification model, consider incorporating additional features alongside the visual embeddings. For example, integrating text descriptions or clinical notes associated with the images can enrich the feature space and improve diagnostic accuracy.
- **Embedding Alignment using CLIP:** If caption descriptions or structured text data are available for the figures, leverage contrastive learning models like CLIP (Contrastive Language-Image Pretraining) to align the image and text embeddings. This multimodal alignment forces semantic separation based on text logic, which can effectively mitigate the overlapping clusters observed in the current embedding space and drastically improve class separability.